

REPUBLIC OF CAMEROON

Peace – Work – Fatherland

MINISTRY OF SECONDARY EDUCATION

INSPECTORATE GENERAL OF EDUCATION

Inspectorate of Pedagogy in charge of the
Teaching of Computer Science



REPUBLIQUE DU CAMEROUN

Paix – Travail – Patrie

MINISTRE DES ENSEIGNEMENTS SECONDAIRES

INSPECTION GENERALE DES ENSEIGNEMENTS

Inspection de Pédagogie chargée de
l'enseignement de l'informatique

NATIONAL HARMONISED PROGRESSION SHEET FOR COMPUTER SCIENCE LOWER SIXTH

SCHOOL YEAR:

SCHOOL:

WEEKLY WORKLOAD: 8 periods

TEACHER:

GRADE:

TELEPHONE:

Term	Week	Module	Category of Action	Lesson No	Lesson Title	Objectives	Resources	Th	Pra	Dig	
	1		Exploring AI Concepts and Emerging technologies	1	Introduction to artificial intelligence – Part 1	• Define AI and explain its history and evolution • Identify and describe the main types of AI. • Explain common applications of AI.					
				2	Introduction to artificial intelligence – Part 2	• Explain ethical issues associated with AI. • Examine the potential risks and benefits of AI. • Discuss principles for the responsible use of AI					
				3	AI techniques and Machine Learning	• Describe common AI techniques and their applications. • Explain the characteristics of intelligent systems. • Compare AI techniques and intelligent systems based on their applications • Define Machine Learning. • Explain the basic principles of Machine Learning. • Distinguish between supervised, unsupervised and reinforcement learning					
				4	Developing AI Systems and the importance of data in AI systems – Part 1	• Explain the stages involved in developing an AI system. • Identify programming languages and tools used in AI development. • Explain why Python is widely used in AI development.					
	2			5	Developing AI Systems and the importance of data in AI systems – Part 2	• Distinguish between primary and secondary data • Classify different types of data including quantitative, qualitative, structured and unstructured data • Relate the quality and type of training data to the performance and reliability of an AI system.					
				6	Data collection and Data analysis	• Use questionnaires, interviews and observation collection methods Collect data using digital forms and online surveys • Describe data analysis					

		Module 1: Computer Applications , Artificial Intelligence & Emerging technologies and their Socio-Economic Implications				- List examples of tools that can be used for data analysis (machine learning, spreadsheets, DBMS.....) - Explain the objectives of data analysis (organise, calculate, transform, interpret....)				
				7	Integration Activity					
			Description of Computer Systems	8	Components of a computer and basic operations. – Part 1	- Describe types of computers - Describe how the computer systems function - Identify input and output devices - Connect correctly input, processing and output devices - Power a computer using soft and warm booting				
				9	Components of a computer and basic operations. – Part 2	- Transfer data/information to/from peripheral devices - Enter information into a computer system - Print a document - Identify the hardware resources commonly required to run AI-enabled applications and explain their roles.				
				10	Integration Activity					
			Utilization of digital resources	11	Communication and collaboration tools	- Describe communication and collaboration tools - Select a suitable communication and collaboration tool for a given situation - Set up a learning environment using communication and collaboration tools - Share information/data over a network - Participate in learning using communication and collaboration tools - Communicate, collaborate and share resources using social media				
				12	Digital systems: manual, automated, processing, simulation and modelling – Part 1	- Describe manual and automated systems - Compare manual and automated systems - Describe monitoring and control systems - Describe batch and online processing systems				
				13	Digital systems: manual, automated, processing, simulation and modelling – Part 2	- Describe simulations and modelling systems - Outline the advantages and disadvantages of manual and automated systems - Explain how automated systems can use AI techniques to monitor conditions, make predictions and support decisions.				
				14	Integration Activity					
		4	Application areas of computers and evaluating the Implications of Computers	15	Computer Applications and Productivity Tools	- Identify domains of use of general-purpose and other computing applications - Exploit productivity tools such as word processors and desktop publishers - Determine the use of general-purpose applications in commerce, industry, science, education, arts and media				

						Accomplish tasks using productivity tools such as spreadsheets, word processors, databases and presentation software				
				16	Social and Economic Implications of Computers, Security of Computer Systems. – Part 1	- Identify social and economic effects of computerisation. - Explain how computing affects employment, products, services, and working environments. - Evaluate the advantages and disadvantages of computerisation to people, organisations, and society. - Explain the concepts of computer system security, reliability, and resilience.				
				17	Social and Economic Implications of Computers, Security of Computer Systems. – Part 2	- Explain how confidentiality, integrity, privacy and availability contribute to computer system security. - Analyse the consequences of system failure, data loss, and security breaches - Evaluate the social, economic, privacy and security implications of deploying AI-based systems.				
				18	Computer Crimes, Professional Ethics and Conduct.	- Define computer crime and identify common types of computer crime. - Explain how selected computer crimes are committed and their effects. - Describe measures used to prevent and combat computer crime. - Identify professional, ethical, and moral responsibilities in the use of information systems. - Explain ethical principles governing the use and management of computerised information. - Apply professional codes of conduct to given computing situations				
					Evaluation					
					Remediation					
				19	Legislation and Global Communication, Responsible and Ethical Use of Computers – Part 1	- Explain the purpose of laws relating to computer misuse, copyright, and data protection. - Analyse the effects of global communication on citizenship and cultural issues. - Evaluate the impact of the digital divide and promote responsible participation in the digital society				
				20	Legislation and Global Communication, Responsible and Ethical Use of Computers – Part 2	- Examine the effects of the use of computer material on health and the workplace - Differentiate between legislation and ethical issues related to the use of computers - Emulate professional, ethical and moral obligations of computing				
				21	Integration Activity					
			Representing data in	22	Computer Hardware Components and	Connect computer hardware components and outline the benefits and drawbacks of embedded systems				

		Computer Organisation and Architecture	Computer Systems		Embedded Systems – Part 1	Describe principal operations of peripherals including laser/3D printers, microphones, speakers, magnetic disks, flash memory, optical drives, touch screens and virtual headsets				
	7			23	Computer Hardware Components and Embedded Systems – Part 2	Explain the roles of internal components including motherboard, power supply, video card and sound card, and relate hardware devices to applications and people				
				24	Base Conversions	- Describe common number systems and their characteristics. - Explain the applications of number systems in computing. - Determine the positional value of digits in different number systems Convert numbers between binary and decimal. Convert numbers between binary, octal and hexadecimal. Convert numbers between decimal, octal and hexadecimal				
				25	operations on Number Systems and Computer Data Units – Part 1	- Carry out arithmetic operations in bases 2, 8, 10 and 16 - Distinguish kibi/kilo, mebi/mega, gibi/giga and tebi/tera				
				26	operations on Number Systems and Computer Data Units – Part 2	- Describe practical applications of BCD and hexadecimal - Explain how binary and hexadecimal representations support the storage and processing of data used by AI systems.				
	8		27	Representation of signed integers	- Describe methods used to represent signed integers, including sign magnitude, one’s complement, and two’s complement. - Represent and interpret signed integers using different binary representation methods					
			28	Signed Integer Arithmetic	- Perform addition and subtraction of signed integers using sign-magnitude, one’s complement, and two’s complement representations. - Identify overflow and underflow conditions when performing signed integer arithmetic					
			29	Representation of characters, images and sound – Part 1	- Explain different character encoding schemes (ASCII, EBCDIC, Unicode) - Describe basic principles of digitizing images (pixels, resolution, colour depth)					
			30	Representation of characters, images and sound – Part 2	- Describe basic principles of digitizing sound (sampling rate, bit depth) - Explain how character, image and sound data can be represented as digital inputs for AI applications.					
	9		31	Integration Activity						
			Designing and Analysing Digital Logic Circuits	32	Digital Logic and Logic Gates	- Explain the role of digital logic circuits in computer systems. - Describe the functions and symbols of basic, universal, and exclusive logic gates. - Construct truth tables for different logic gates				
				33	Logic Circuit Design 1 – Part 1	- Define combinational logic circuit - Explain the relationship between logic gates, inputs, and outputs in digital circuits.				

						• Design simple combinational circuits using logic gates.				
				34	Logic Circuit Design 1 – Part 2	• Analyse digital circuits by deriving outputs and constructing truth tables • Explain how logic circuits provide the basic computational operations used by AI-enabled digital systems.				
				35	Binary Arithmetic Circuits	• Explain the operation of half-adder and full-adder circuits. • Design and analyse binary addition circuits using logic gates. • Explain the operation of half-subtractor and full-subtractor circuits				
				36	Sequential Logic Circuits	• Define sequential logic circuit • Explain the difference between combinational and sequential logic circuits. • Describe the purpose and operation of flip-flops in digital systems. • Derive truth tables and determine outputs of SR and JK flip-flops				
				37	Practical Digital Logic and Sequential Circuits	• Construct logic circuits/truth tables, implement universal NAND/NOR logic, design simple circuits such as traffic lights, deduce inputs/outputs and identify application areas • Produce truth tables for half/full adders; describe SR and JK flip-flops and their role as storage elements; derive flip-flop truth tables • Perform Boolean algebra using De Morgan's laws • Relate sequential logic and storage elements to the operation of hardware used in AI and intelligent systems.				
				38	Integration Activity					
					Evaluation					
					Remediation					
				39	Introduction to Computer Organisation and Architecture – Part 1	• Define computer organisation and computer architecture. • Differentiate between organisation and architecture. • Identify major computer system components.				
				40	Introduction to Computer Organisation and Architecture – Part 2	• Explain the roles of the CPU, memory and I/O devices. • Explain the stored-program concept • Describe and illustrate the organisation of Von Neumann and Harvard architectures				
				41	Advanced Processor Architectures – Part 1	• Outline differences between RISC and CISC architectures and the importance/use of pipelining and registers in RISC • Compare Von Neumann with Flynn(SISD, SIMD, MISD, MIMD), CRAY, Dataflow and Harvard architectures • Analyse parallel processing techniques used to improve computational performance				
				42	Advanced Processor Architectures – Part 2	• Outline roles of virtual machines and explain their benefits and limitations. • Compare the advantages and limitations of Von Neumann and Harvard architectures				

						• Explain why parallel processing and specialized processor architectures are important for AI workloads.				
			Exploring Computer Organisation and Architecture	43	System Communication and Timing	• Explain the purpose of system buses in a computer system. • Describe the functions of the data bus, address bus, and control bus. • Illustrate how system buses enable communication between the CPU, memory, and I/O devices • Explain how data are transferred between computer-system components • Explain roles of the system clock and Immediate Access Store (IAS)				
				44	The Processor, Its Components and the machine cycle	• Explain the functions of the major components of the processor, including the Control Unit, ALU, and registers. • Describe fetch, decode and execute stages. • Trace instruction execution • Identify registers used				
				45	Interrupts, Polling and Parallel Processing	• Explain situations in which polling occurs and where interrupts occur; compare pipelining and parallel processing • Describe interrupt-detection strategy. • Relate cycle efficiency to performance • Explain how interrupts and parallel processing can improve the responsiveness and performance of AI-enabled systems.				
				46	Assembly Instructions and Addressing Modes	• Trace a simple assembly-language program and explain/manipulate instruction sets, including data movement, I/O, arithmetic, unconditional/conditional and compare instructions • Give examples of instruction formats; explain fixed- versus variable-length instructions; and distinguish immediate, direct, indirect, register, relative and indexed addressing modes				
				47	Integration Activity					
		13		48	Primary Storage – Part 1	• Explain the concept of memory hierarchy. • Evaluate the relationship between speed, capacity, and cost in memory organisation • Explain the functions and characteristics of primary memory (RAM, ROM, Cache). • Differentiate between RAM and ROM types.				
				49	Primary Storage – Part 2	• State applications of RAM, ROM and Cache • Differentiate SRAM and DRAM • Explain how memory capacity, speed and hierarchy affect the performance of AI applications and data processing.				
Seco nd Term		14	Managing Computer Memory	50	Memory Addressing Concepts	• Explain memory addressing concepts. • Perform memory capacity calculations. • Calculate the number of memory locations from address lines. Solve problems involving memory organisation				
				51	Secondary Storage Technologies	• Describe different secondary storage technologies (HDD, SSD, optical media).				

						- Classify storage devices based on their technologies. - Analyse factors that influence the performance of storage devices				
				52	Integration Activity	- Integrate the concepts and practical skills developed in the category to solve a contextual problem. - Apply memory concepts to propose an appropriate storage arrangement for an AI application and justify the choice.				
				53	Peripheral Devices and Interfaces	- Analyse the characteristics used to evaluate the performance and suitability of peripheral devices. - Explain how different interfaces and communication methods that enable peripherals to exchange data with a computer system				
				54	The I/O Controller	- Explain the role of I/O controllers in managing communication between the processor and peripheral devices. - Describe polling and interrupt-driven processing as techniques for handling I/O operations				
				55	Integration Activity					
					Evaluation					
					Remediation					
				56	Algorithm Design Approaches 1	Explain common algorithm design approaches/strategies (brute force, divide and conquer, stepwise refinement, greedy algorithm,)				
				57	Algorithm Design Approaches 2	- Differentiate between top-down and bottom-up design. - Apply stepwise refinement to develop an algorithm				
				58	Representation of algorithms – Part 1	- Describe forms and characteristics of algorithms and write steps to solve a problem - Distinguish good and poor algorithms with respect to stated characteristics - Use control constructs to show how a problem can be solved and determine the appropriate course of action under prevailing conditions				
				59	Representation of algorithms – Part 2	- Interpret algorithms represented in various ways, convert an algorithm from one form to another and represent an algorithm in various ways - Use an algorithmic representation to describe a simple AI decision-making or classification process.				
				60	Algorithm Design Strategies	- Identify the strategies used in existing algorithms. - Select an appropriate strategy for a given problem				
				61	Sequence and Selection Structures 1	- Explain the concept and need for control structures - Explain sequence and selection control structures				
				62	Sequence and Selection Structures 2	- Define dry running and trace table - Design and trace algorithms using simple selection (decision-making) constructs - Design and trace a simple decision algorithm that could be used as a rule-based component of an AI system.				

	18			63	Sequence and Selection Structures	• Explain the need for multiple selection structures in algorithm design. • Design algorithms using nested IF and CASE constructs. • Trace algorithms involving multiple decision paths				
				64	Repetition Structures 1a	• Explain the concept, purpose, and types of repetition structures. • Define count controlled loop • Explain the syntax, characteristics, and execution of the FOR loop. • Construct and trace algorithms using the FOR loop to solve simple computational problems				
				65	Repetition Structures – Part 1	• Explain the concept and characteristics of condition-controlled loop. • Distinguish between post test and pre-test loop • Describe the syntax and execution of the WHILE and REPEAT...UNTIL loops.				
				66	Repetition Structures – Part 2	• Construct and trace simple algorithms using the WHILE and REPEAT...UNTIL loops • Develop a repetition-based algorithm for processing a dataset as a preliminary step in an AI application.				
	19			67	Subroutines	• Explain the concept, purpose, and characteristics of subroutines in algorithm design. • Differentiate between procedures and functions. Demonstrate how subroutines are called within algorithms				
				68	Parameter Passing	• Explain the purpose of parameters in subroutines. • Differentiate between formal and actual parameters. • Explain different methods of passing parameters to subroutines, including pass-by-value and pass-by-reference				
				69	Processing Subroutine Calls	• Explain how the processor executes subroutine calls during program execution. • Describe the role of the stack in storing return addresses and supporting subroutine execution • Explain how subroutines can modularise data-processing tasks within an AI-oriented program.				
				70	Processing Subroutine Calls	• Trace the sequence of operations involved in calling and returning from a subroutine. • Explain the concept and characteristics of recursion				
	20			71	Recursion	• Define recursion • Trace the execution of recursive algorithms. • Develop simple recursive solutions to computational problems				
				72	Recursion Limitations and Complexity	• Explain how the operating system uses stacks during recursive function calls. • Differentiate between iteration and recursion Dry run recursive algorithms • Assess limitations of recursion, including high runtime space use and difficulties with high values				

			Designing Modular Algorithms			• Compare iterative and recursive approaches for solving computational tasks that may occur in AI data processing.					
				73	Integration Activity						
					Evaluation						
	21	Programmin g I	Understanding Programming Language Concepts		Remediation						
				74	Programming Language Basics	• Explain the role of programming languages (PLs) in software development. • Categorise PLs into low-level and high-level languages. • Compare the characteristics, advantages, and limitations of low-level and high-level PLs					
				75	programming paradigms 1	• Define programming paradigm • Describe imperative(procedural, OOP) and declarative (Logic, functional, SQL) programming paradigms. • Differentiate the imperative and declarative programming paradigms using suitable examples. • Justify the use of imperative and declarative programming for different software development tasks • Compare imperative and declarative programming approaches used in the development of AI and data-driven applications.					
				76	programming paradigms 2	• Explain the core concepts of object-oriented programming (OOP): class, object, attributes, and methods, using suitable examples. • Describe, with suitable examples, the key characteristics of OOP: encapsulation, abstraction, inheritance (single, multiple, and multi-level), polymorphism, coupling, and cohesion					
				77	Language Translators	• Explain the purpose of language translators. • Describe and distinguish the functions of an assembler, compiler, and interpreter. • Explain the main stages involved in program translation and compilation					
				22	78	Program Errors and Debugging	• Explain the terms syntax, semantics, syntax error, semantic error, and runtime error, using suitable examples. • Describe the purpose and use of common debugging concepts and tools, including debuggers, breakpoints, watchpoints, and step execution • Use debugging tools to identify and correct errors in a program that processes data for an AI application.				
					79	Integration Activity					
					80	Introduction to C & Python Programming 1	• Describe the features and applications of C & Python. • Explain the basic syntax and structure of C & Python programs				
					23	81	Introduction to C & Python Programming 2	• Write simple C & Python programs using print statement - Declare and use variables in C & Python • Write a simple program that prepares or formats input data for use in an AI application.			

				82	Introduction to C & Python Programming 3	Identify and use C & Python data types (integer, float, string, Boolean)				
				83	Python/C Operators and Expressions 1	- Identify common C & Python operators. - Explain how C & Python expressions are evaluated. - Construct C & Python expressions to solve simple problems				
				84	Python/C Operators and Expressions 2	- Use input() in Python or scanf() in C to accept user input - Combine input and output statements in simple programs - Use program input and output to collect, validate and display data that could be used by an AI system.				
				85	Python/C Selection Structures 1	- Explain the use of selection structures in C & Python. - Construct C & Python programs using simple and multiple selection				
				86	Python /C Selection Structures 2	Apply selection structures to solve computational problems C & Python				
				87	Python/C Iterative Structures	- Explain the use of iteration structures in C & Python. - Construct C & Python programs using for and while loops - Apply iteration structures to solve computational problems - Apply iteration structures to process a dataset or repeated computations used in a simple AI-related task.				
Third Term				88	Writing and Debugging C programs	- Apply algorithmic design C coding to solve a simple problem - Debug simple syntax and logic errors in C programs				
				89	Writing and Debugging Python programs	- Apply algorithmic design Python coding to solve a simple problem - Debug simple syntax and logic errors in Python programs				
				90	Python Program Development and Debugging Project – Part 1	- Apply algorithmic design to develop a complete Python solution to a contextual problem. - Test and debug the program using systematic techniques.				
				91	Python Program Development and Debugging Project – Part 2	- Present and explain the final solution. - Develop, test and explain a Python solution for a contextual problem involving data or a simple AI-related feature.				
				92	Integration Activity					
					Evaluation					
					Remediation					
				93	Primitive Data Types	- Explain the concept and characteristics of primitive data types used in programming. - Describe common primitive data types and their applications. - Explain the need for data structures in organising and managing collections of data				
				94	Arrays	- Explain the structure and characteristics of arrays. - Distinguish between 1D and 2D arrays. - Represent and manipulate data using arrays				

						• Use arrays to organise and process a collection of data that could serve as input to an AI application.				
	27			95	Records	• Explain the structure and characteristics of records. • Differentiate between arrays and 2D records based on their organisation and use. • Represent and manipulate data using records				
				96	Data Types, Memory Allocation and Data Structures	• Associate a data type with the amount of memory allocated to an identifier having that data type • Calculate the memory allocated to each simple data type • Represent a range of data of the same type as an array; declare string data as an array of characters; represent heterogeneous data as a record; and declare a list (enumeration) • Compare pointer-driven data structures with fixed data structures such as arrays				
				97	Linked lists	• Explain the structure and characteristics of linked lists. • Represent and manipulate data using linked lists in programming problems • Explain how linked lists can support dynamic data management in applications that process AI-related datasets.				
				98	Hash Tables	• Explain the structure and characteristics of hash tables. • Represent and manipulate data using hash tables in programming problems. • Explain linear probing and its importance when managing collision in hash tables • Explain how linked lists can be used to handle collisions in hash tables through separate chaining				
	28			99	Using Repetition Structures to process arrays and lists	• Develop algorithms using repetition structures to process arrays, validate input, generate tables, and print repetitive patterns				
				100	Integration Activity	• Integrate the concepts and practical skills developed in the category to solve a contextual problem. • Integrate data-structure skills to design an efficient data-processing solution for a simple AI application.				
		Algorithm Efficiency		101	Complexity of Algorithms	• Explain the concepts of time complexity and space complexity. Estimate the time and space requirements of simple algorithms by counting operations and memory usage. • Compare simple algorithms based on their estimated resource requirements				
				102	Big-O Notation	• Explain the purpose of Big-O notation in algorithm analysis. • Express the time complexity of simple algorithms using Big-O notation. • Compare the time complexity of common searching and sorting algorithms • Explain the concepts of P and NP complexity classes.				

	29		Evaluating Algorithm Efficiency	103	Computational problem types	• Distinguish between tractable and intractable problems using appropriate examples • Differentiate between decidable and undecidable problems (Halting problem, Travelling Salesman problem) • Provide an informal definition of turing machine. • Differentiate between computable and non computable functions using appropriate examples • Explain how heuristics can be applied to obtain practical solutions to intractable problems. • Explain why some AI-related computational problems are computationally difficult and relate them to P and NP concepts.				
				104	Algorithm Complexity, Dry Runs and Optimisation	• Determine time and space complexity from the number of executions of key operations and perform dry runs • Optimize an algorithm to improve efficiency and evaluate efficiency in terms of number of steps				
				105	Integration Activity					
				106	Introduction to Computer Networks 1	• Define Computer Network. • Identify and give the role of network devices. • Explore the main network types by geographic scope • Identify network components and explain how network connectivity enables AI services and cloud-based intelligent applications.				
	30	Computer Networks and Data Communication	Exploring Internet Technologies	107	Introduction to Computer Networks 2	• Distinguish between peer-to-peer and client-server network. • Describe star, bus, ring and mesh topologies. • Describe the different types of network cabling7media. • Classify transmission media as guided or unguided				
				108	Introduction to Computer Networks 4	• Compare common transmission media based on their characteristics. • Select suitable transmission media for given applications				
				109	Network Connections, Architectures and Hardware	• Describe types of network connections such as point-to-point and multipoint • Explain features of network operating systems, including multitasking and multi-user features • Outline network software • Differentiate types of network architectures • Select and connect network hardware components such as cables, switches, routers and modems • Compare transmission media using data rate, transmission distance and ease of installation • Explain how network architecture and hardware support the transmission of data to and from AI and cloud services.				
				110	The OSI model	• Explain the purpose of communication protocols and standards. • Describe the layers of the OSI reference model. • Relate communication protocols to the layers of the OSI model				

	31			111	IP addressing	• Explain the Structure of an IP address. • Explain the different Classes of IP address. • Describe the different types of IP address. • Explain the notion of CIDR and subnetting				
				112	Network Configuration and Troubleshooting	• Configure the operating system for networking • Explain how mobile communication technology works • Explain the difference between wired and wireless transmission • Outline errors that may occur in a network • Set up and troubleshoot a network • Use network troubleshooting techniques to maintain reliable connectivity for an AI-enabled or cloud-based application.				
				113	Data Communication Modes and Transmission	• Identify the features of data-communication networks • Describe simplex, half-duplex and duplex communication modes • Differentiate parallel and serial digital-data transmission • Explain synchronous and asynchronous transmission • Explain the functions of multiplexers and demultiplexers • Differentiate baseband and broadband transmission • Explain the purpose of data transmission and determine differences between transmission modes • Compare serial and parallel transmission based on speed and number of bits transmitted at a time • Explain advantages and disadvantages of analogue and digital signal transmission				
				114	Network and Data Security	• Explain data security and safety principles for protecting data/networks from malware and unauthorized access • Use antivirus software to protect a computer/network from viruses, Trojan horses and worms • Describe error-detecting codes such as parity bits, Hamming codes and cyclic redundancy checks/checksum • Explain measures used to protect computers and networks from intruders and natural disasters, including passwords, firewalls, encryption and backup • Scan a computer system using antivirus software • Set up a firewall and web filtering using antivirus software • Save data on cloud-storage systems such as Google Drive				
	32			115	Network Standards, Protocols and Internet Development	• Describe different network standards and protocols • Produce a report comparing the OSI and TCP reference models • Discuss Internet protocols such as TCP, UDP, IP and FTP • Describe the history of the Internet • Explain intranet, extranet and Internet • Explain how Internet protocols support the exchange of data required by AI and cloud-based services.				
				116	Internet Services and Safe Online Use	• Describe Internet services such as e-commerce, e-learning and e-banking • Exchange information using email and use search engines				

						• Do business online while observing safety and security risks • Change privacy settings and ensure privacy online					
				117	Integration Activity	• Integrate the concepts and practical skills developed in the category to solve a contextual problem. • Evaluate the safe and responsible use of AI-enabled Internet services, including privacy, security and information reliability.					
			Introductio n to Web Developme nt Technologie s	Designing simple Web Pages	118	Introduction to Web technology, Internet Connectivity, Online Services and Information Literacy	• Describe common Internet services and their uses. • Explain the functions of web browsers, search engines, and email systems. • Explain the role of markup languages in web development • Select suitable hardware/software for Internet connectivity and access websites using a browser • Select an appropriate Internet service for a given context and manage browser bookmarks • Apply for a job or scholarship online • Assess online information for relevance, bias, validity, reliability and sufficiency • Download images, files, software and drivers from the Internet				
	33	119			Introduction to HTML 1	• Introduce the structure of an HTML document. • Use basic HTML tags for text and headings • Use HTML tags for lists, links, and images.					
		120			Introduction to HTML 2	• Use HTML tags to create tables • Assess AI-generated or AI-assisted online information for relevance, bias, validity, reliability and sufficiency.					
		121			Introduction to CSS 1	• Introduce CSS and its role in styling • Apply CSS to style text, colours, and layout. Combine HTML and CSS to build a structured webpage					
		122			Integration Activity						
					Evaluation						
					Remediation						
	34	Revision									
	35										
	36										